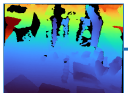


Inputs

intrinsics $\mathbf{K}$



keyframe $\mathcal{I}^k$ and
frame $\mathcal{I}^f$



sensor depth $\mathbf{Z}$

FRONT END every frame

1

Depth-conditioned Pow3R

two-view pointmaps and confidences
dense even where
the sensor has holes

cond

2

Metric scale

from the
sensor depth

scale

3

Tracking and keyframing

Sim(3) pose from projective matches
new keyframe when overlap drops

new keyframe

lost

relocalized

5

Depth-anchored backend

keyframe geometry held
to the sensor depth
all keyframe poses refined together

anchor

4

Loop closure, relocalization

retrieval on Pow3R's tokens
verified by projective matching

edges

BACKEND every new keyframe

final poses

Outputs

metric trajectory



estimate vs. ground truth

dense metric map



keyframes + map-only keyframes